PROCESS 2 – ESSAY

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Ths. Nguyen Thanh An

***Tóm tắt nội dung*—Summarize**

Requirements:This task develops an inference system based on propositional logic using the Forward Chaining algorithm.The system allows users to define a Knowledge Base (KB) with logical clauses, and then infer whether a particular query logically follows from the KB.

Approaches:

- **Representing rules as Clause objects with premises and conclusions.**
- **Managing all clauses and symbols via the Problem class.**
- **Applying the Forward Chaining algorithm to perform automated reasoning.**
- **Visualizing logical relationships using Graphviz.**

Experimental results:

- **Verify entailment**

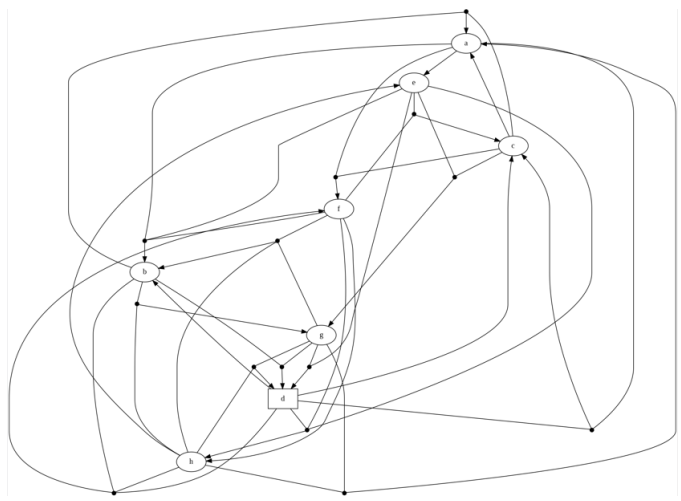

```
Please enter a character: c
Query: c

Is 'c' entailed by the knowledge base? True
Query result: True

Inference trace:
Step 1: Processing symbol 'd'
  Decremented count for clause 'd => b' to 0
  All premises satisfied, adding 'b' to agenda
  Decremented count for clause 'b ∧ d ∧ h => f' to 2
  Decremented count for clause 'd => c' to 0
  All premises satisfied, adding 'c' to agenda
  Decremented count for clause 'd ∧ e ∧ f => h' to 2
  Decremented count for clause 'a ∧ d => c' to 1
Step 2: Processing symbol 'b'
  Decremented count for clause 'b ∧ c => a' to 1
  Decremented count for clause 'b ∧ d ∧ h => f' to 1
  Decremented count for clause 'b ∧ g => d' to 1
  Decremented count for clause 'b ∧ h => g' to 1
Step 3: Found query 'c' in agenda

Inferred symbols in order:
d, b
```

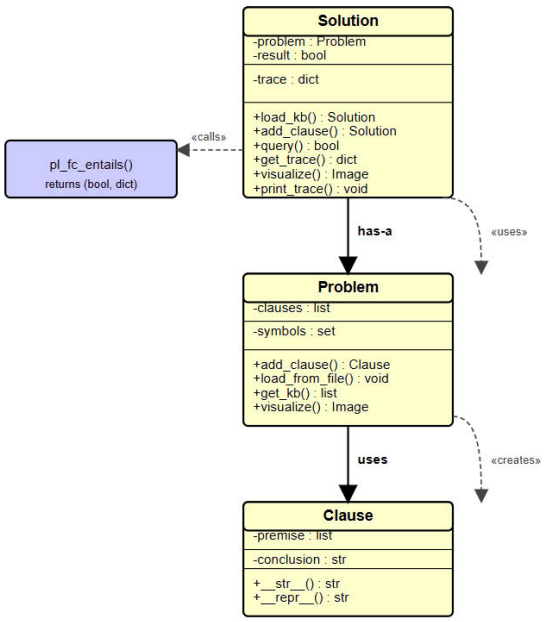
- **Graph**



Levels of completion: 87%

I. Task 1

A. Class Diagram



B. Clause Representation

A Clause class is defined to represent individual inference rules. Each clause comprises a list of premises and a single conclusion. The clause is formatted as "premise$_1$ $\wedge$ premise$_2$ $\wedge \ldots \Rightarrow$ conclusion". If the premise list is empty, the output defaults to "$\top \Rightarrow$ conclusion".

Clause	
Attribute	- premise: list - conclusion: str
Method	+ __init__(premise, conclusion) + __str__(): str + __repr__(): return __str__()

C. Problem Representation

The Problem class is responsible for storing the entire knowledge base (KB), managing all clauses and symbols, and providing functionality for visualizing logical relationships.

Problem	
Attribute	- clauses: list - symbols: set
Method	+ __init__(kb_file=None) + add_clause(premise, conclusion): Clause + load_from_file(filename): None + get_kb(): list + visualize(output_file): Image

D. Implementing Forward Chaining Algorithm

We implement the Forward Chaining algorithm as a standalone function named pl_fc_entails. It starts with known facts (clauses without premises) and repeatedly applies rules whose premises are satisfied until either:

- the query is entailed (success), or
- no new inference can be made (failure).

```
function PL-FC-ENTAILS?(KB, q) returns true or false
inputs:
  KB - the knowledge base, a set of propositional definite clauses
  q - the query, a proposition symbol

  count ← a table, where count[c] is the number of symbols in c's
   premise
  inferred ← a table, where inferred[s] is initially false for all symbols
  agenda ← a queue of symbols, initially symbols known to be true in KB

while agenda is not empty do:
  p ← POP(agenda)
  if p = q then return true
  if inferred[p] = false then:
    inferred[p] ← true
    for each clause c in KB where p is in c.PREMISE do:
      decrement count[c]
      if count[c] = 0 then:
        add c.CONCLUSION to agenda

return false
```

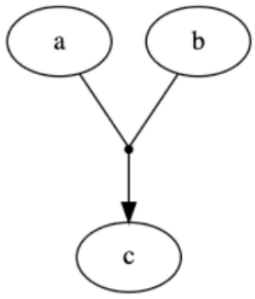
E. Solution Wrapper

A Solution class wraps everything: loading KB from file, querying, printing traces, and visualizing. This simplifies user interaction with the system and supports interactive testing.

Solution	
Attribute	- problem: Problem - result: bool - trace: dict
Method	+ __init__(kb_file=None) + load_knowledge_base(kb_file): Solution + add_clause(premise, conclusion): Solution + query(query_symbol): bool + get_trace(): dict + visualize(output_file): Image + print_trace(): None

F. Visualizing Logical Clauses with Graphviz

Example :
input : -a -b c
ouput:



G. Result

input : data.txt
output :

- Verify entailment

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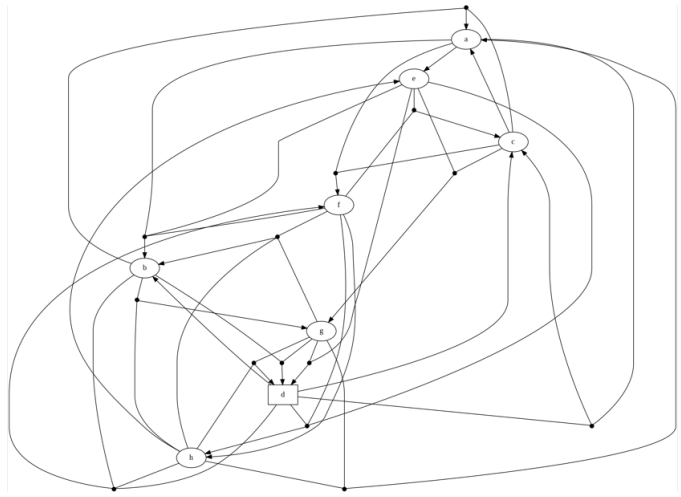
- Graph



II. Contributions

Name	Task Description	Completion
Nguyen Dong Quan	Make report and support task 1	90%
Pham Quoc Hung	Do Task 1	90%
Dinh Bui Khanh Huy	Do Task 1	90%

III. Self-evaluation

Task	Score
Task 1	7.0
Report	1.0

IV. Conclusion

Requirements:This task develops an inference system based on propositional logic using the Forward Chaining algorithm.The system allows users to define a Knowledge Base (KB) with logical clauses, and then infer whether a particular query logically follows from the KB.

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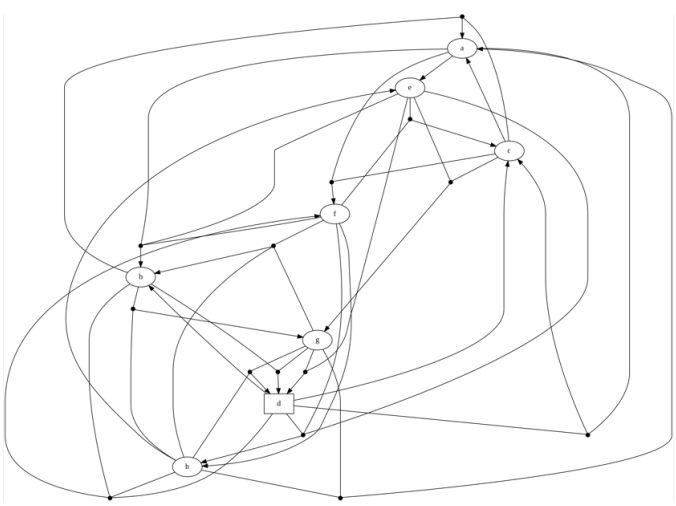

Levels of completion: 87%

References

[1] Stuart Russell and Peter Norvig. 2009. Artificial Intelligence: A Modern Approach (3rd ed.). Prentice Hall Press, Upper Saddle River, NJ, USA.

[2] Lê Hoài Bắc, Tô Hoài Việt. 2014. Giáo trình Cơ sở Trí tuệ nhân tạo. Khoa Công nghệ Thông tin. Trường ĐH Khoa học Tự nhiên, ĐHQG-HCM.

[3] Nguyễn Ngọc Thảo, Nguyễn Hải Minh. 2020. Bài giảng Cơ sở Trí tuệ Nhân tạo. Khoa Công nghệ Thông tin. Trường ĐH Khoa học Tự nhiên, ĐHQG-HCM.